

§27 — Can TerraMind embeddings resolve sub-kilometre soil-moisture contrast?

Design document, written before any code exists, for critique

Soil moisture project

2026-08-12

Abstract

Six TxSON stations inside one 2.24 km tile have observed mean soil moisture spanning 0.137–0.287 m³/m³. The model predicts them at 19% of that spread, with the station ordering *anti*-correlated with truth. This document specifies an experiment that asks one question — do the frozen TerraMind token embeddings contain information that separates two stations a few hundred metres apart? — and derives the mathematics that makes the answer identifiable. Nothing here has been run. The purpose is to be criticised before compute is spent.

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1 The observation

1.1 What the tile figure shows

Figure 1 is `figures/tile_context/ISMN-TxSON-CR200-18.pdf`. Six TxSON stations fall inside one 224×224 @ 10 m tile (2.24 km), separated by at most 936 m. The tile is not uniform: 45 m of relief, six land-cover classes, one station (CR200-6) in crops in a valley bottom, the rest rangeland on slopes.

- **Observed** across-station spread $0.0601 \text{ m}^3/\text{m}^3$; level range 0.171.
- **Predicted** across-station spread $0.0113 = 19\%$ of observed.
- $r(\text{predicted level, observed level}) = -0.175$.

The same collapse holds on every dense tile: CR200-3 19%, CR200-26 19%, CR1000-2 15% — all below the 0.35 threshold at which `plot_network_timeseries.py` declares “the map repeats ~one series over the whole tile”.

1.2 Two facts that rule out the easy explanations

It is not a readout artefact. One might object that reading six stations off *one* tile’s map is out-of-distribution, since only pixel (112, 112) is ever supervised (`model.py:779`). But reading each station from its **own** centred patch — exactly the training configuration — still collapses: over the 18 OOS TxSON stations, $\text{sd}_{\text{pred}} = 0.0158$ against $\text{sd}_{\text{obs}} = 0.0633$, with $r = -0.11$.

It is not a general failure of level prediction. Skill degrades monotonically with separation and is perfectly healthy at continental range (Table 1, depth 0–10, station-mean level, from `eval_output/predictions_oos.parquet`).

scale	n	sd_{obs}	sd_{pred}	ratio	r
between networks	21	0.0578	0.0525	0.91	+0.85
within USCRN	12	0.0967	0.0851	0.88	+0.89
within SCAN	33	0.1029	0.0868	0.84	+0.78
within SNOTEL	73	0.0715	0.0442	0.62	+0.31
within COSMOS-UK	6	0.0525	0.0356	0.68	−0.05
within TxSON	18	0.0633	0.0158	0.25	−0.11
within one tile	6	0.0601	0.0113	0.19	−0.18

Table 1: Level skill by spatial scale. The model resolves *where is wetter* at ≥ 10 km and not below.

This is consistent with everything measured earlier: §23 found the painted 224^2 structure *anti*-correlated with the landscape ($r(\text{anomaly, DEM}) = -0.04$ to $+0.21$) with 64–83% of variance on the 14×14 token grid; §24.12 found site identity dominates the prediction and its share grows with depth; §22 found “the model emits nearly the same spread at every station”.

1.3 Why the inputs make this the expected outcome

For the six CR200-18 stations, almost the entire input is shared:

- **ERA5-Land (~ 9 km) is byte-identical** for all six (`dataset.py:1045`) and supplies 365 of ~ 600 sequence tokens — numerically dominant in the FiLM context vector (`model.py:743`). SIF (~ 5 km) and TWSA (GRACE) likewise identical.
- **Soil is OpenLandMap 250 m** resampled to 30 m. The six stations differ by 3 % clay, 4 % sand, 1 pH unit — noise.
- **The finest satellite scale the model sees is 320 m, not 160 m.** `dataset.py:212` gives `widths = [1,3,5,7]` $\Rightarrow 2\times 2$ / 6×6 / 10×10 / 14×14 token windows. The `model.py:109` docstring claiming a 1×1 centre scale is wrong. `dataset.py:306` additionally drops 50 % of tokens *before* pooling, which hits the 4-token finest scale hardest.
- **No terrain derivative exists anywhere** — DEM enters only as a pooled TerraMind L12 embedding.
- **No station identity, no per-station bias, no per-station label normalisation;** `dataset.py:1074–1081` emits raw m^3/m^3 , and decoder `BatchNorm2d` (`model.py:176,179`) actively strips per-sample offsets.

A homogeneous prediction is therefore close to the Bayes-optimal answer *given these inputs*. The open question is whether better use of what we already hold could do better.

TxSON_CR200-18 — one 2.24 km tile, 6 soil-moisture stations, 2019-2020

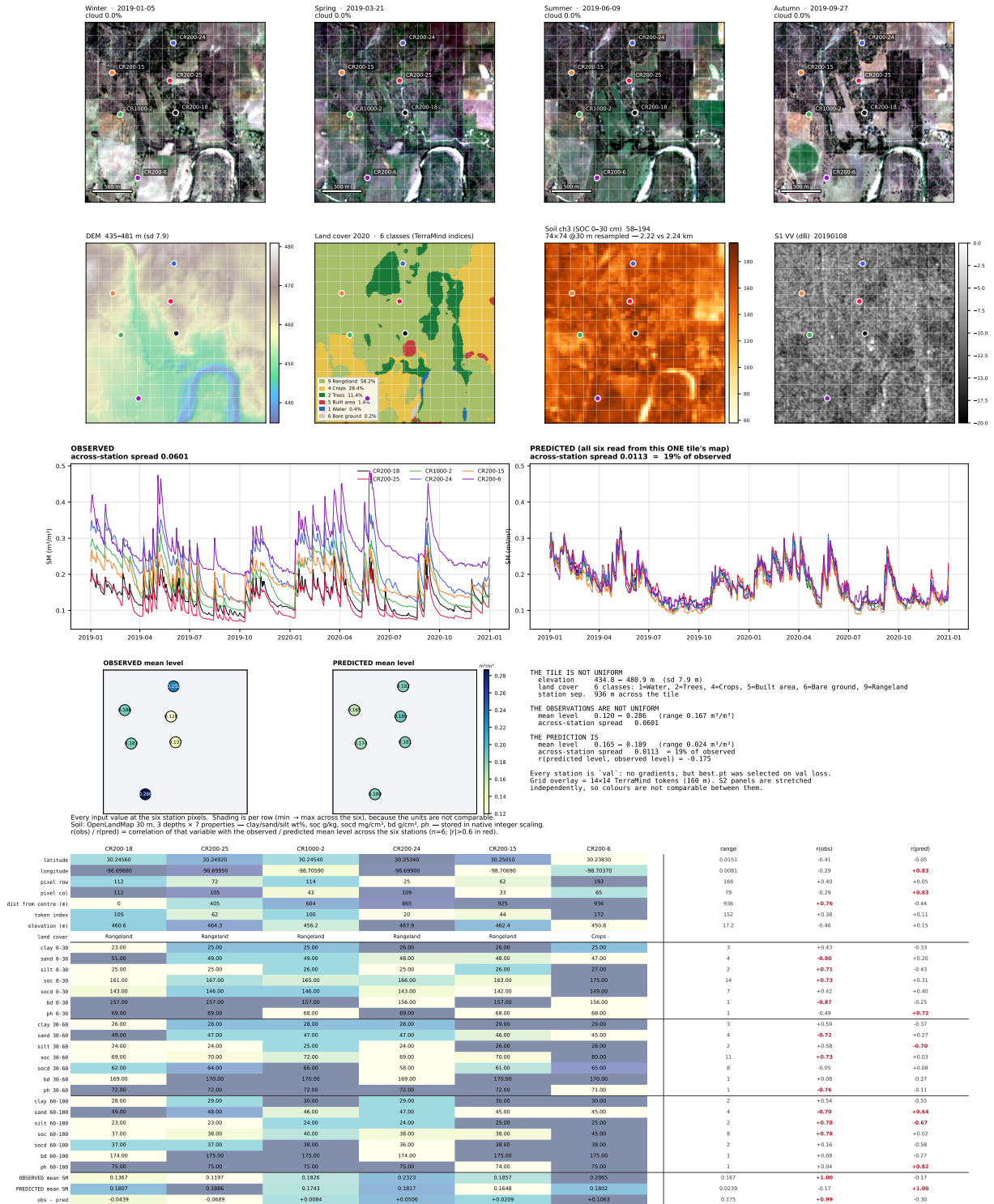


Figure 1: ISMN-TxSON_CR200-18 — one 2.24 km tile, six stations, 2019-2020. Row 3 left: observed series, spread 0.0601. Row 3 right: predicted, spread 0.0113. Row 4: observed vs predicted mean level, and the verdict box.

2 The question

Do the frozen TerraMind token embeddings contain information that distinguishes two soil-moisture stations a few hundred metres apart — and if so, at which layer, for which modality, and does the model actually receive it?

Scope is deliberately narrow. Terrain derivatives, 30 m optics, higher-resolution soil, multi-station supervision and any loss change are *out of scope* and remain deferred. No model, dataset or loss code is modified by this experiment.

One structural advantage of probing the representation rather than the model. The TerraMind encoder is **frozen** and was never trained on soil moisture. Therefore train/val/test leakage does not apply to this experiment at all: a station in the training split is no more “seen” by φ than an out-of-sample one. Every colocated station is usable regardless of split — which matters, because colocated stations are scarce (§6).

3 The experiment in one page

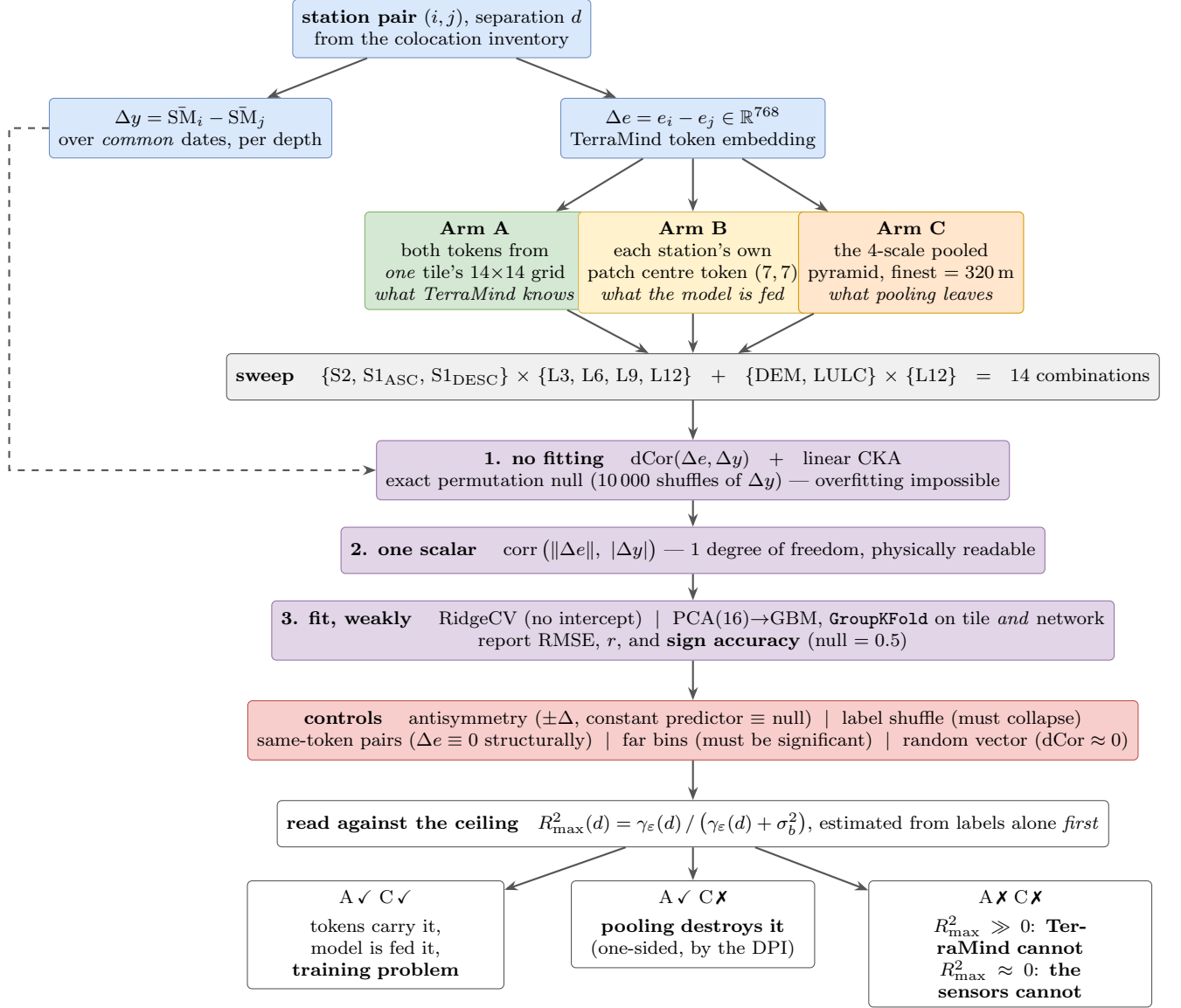


Figure 2: The whole experiment. Blue = data, green/yellow/orange = the three embedding arms, purple = the statistical ladder (least overfittable first), red = controls, and white = interpretation.

4 Mathematics

4.1 Model and estimand

Let station i sit at location x_i and report a long-term mean

$$y_i = \mu(x_i) + b_i, \quad \mathbb{E}[b] = 0, \quad \text{Var}(b) = \sigma_b^2, \quad (1)$$

where $\mu(\cdot)$ is the true mean soil-moisture field and b_i is a *station offset*: probe calibration, installation depth, local bulk density, the mismatch between a point sensor and any areal quantity. Crucially b_i is **not** a property of the location — two probes in the same soil pit would differ.

The embedding is $e_i = \varphi(x_i) \in \mathbb{R}^{768}$, the frozen TerraMind token covering x_i . Decompose the field geostatistically,

$$\mu(x) = m(z(x)) + \varepsilon(x), \quad (2)$$

with z the regional covariates (climate, biome, physiography) varying on a scale $L \approx 100$ km, and ε a zero-mean short-range residual field with variogram $\gamma_\varepsilon(d) = \frac{1}{2}\mathbb{E}[(\varepsilon(x) - \varepsilon(x+d))^2]$.

We want to test whether φ carries information about μ at short range:

$$H_0 : \Delta e \perp \Delta \mu \quad \text{conditional on} \quad \|x_i - x_j\| = d, \quad d \text{ small.} \quad (3)$$

4.2 Why the naive experiment fails, and differencing fixes it

Regressing y_i on e_i across all 993 stations would succeed *for the wrong reason*: L12 encodes climate and biome, so it separates Texas from Norway. High R^2 , zero bearing on separating CR200-18 from CR200-6 at 936 m.

Take differences instead. For a pair at separation $d \ll L$,

$$\Delta \mu = \underbrace{\nabla m \cdot \Delta z}_{\rightarrow 0 \text{ as } d \rightarrow 0} + \Delta \varepsilon \rightarrow \Delta \varepsilon. \quad (4)$$

The pair difference is a high-pass filter. It annihilates everything varying on scales $\gg d$ — and it does so *exactly*, not approximately, for anything constant within the tile. Formally, for any nuisance variable c the contribution to Δy is $c(x_i) - c(x_j)$, which is identically zero when c is tile-constant. That covers: climate, the ERA5-Land grid cell, rainfall history, season and date (we use common dates only), biome, network operator, sensor vendor, calibration epoch — and in Arm A, the entire image context of the TerraMind forward pass, because both tokens come out of *one* forward pass on *one* image.

Hence

$$\text{Cov}(\Delta e, \Delta y) = \underbrace{\text{Cov}(\Delta e, \Delta \varepsilon)}_{\text{the quantity of interest}} + \underbrace{\text{Cov}(\Delta e, \Delta b)}_{= 0 \text{ in expectation}}. \quad (5)$$

The second term vanishes because sensor idiosyncrasy is not visible from orbit: b is a property of the instrument, independent of the land surface that φ encodes.

Identification claim. A statistically significant association between Δe and Δy at small d can only arise from genuine short-range field information in the embedding.

There is no confounder left to explain it away.

This is the entire justification for the pair design, and it requires no modelling assumption — only that $b \perp \varphi$.

4.3 The ceiling: what no method can beat

From (1),

$$\text{Var}(\Delta y) = 2\gamma_\varepsilon(d) + 2\sigma_b^2, \quad (6)$$

so the maximum variance in Δy that *any* function of location can explain is

$$R_{\max}^2(d) = \frac{\gamma_\varepsilon(d)}{\gamma_\varepsilon(d) + \sigma_b^2} = \frac{\text{SNR}(d)}{1 + \text{SNR}(d)}, \quad \text{SNR}(d) := \frac{\gamma_\varepsilon(d)}{\sigma_b^2}. \quad (7)$$

As $d \rightarrow 0$, $\gamma_\varepsilon(d) \rightarrow 0$ while σ_b^2 does not. **Below some separation, no covariate whatsoever can predict Δy** , because what remains is pure instrument noise. This is a property of the observing system, not of the model.

σ_b^2 is estimable: it is the **nugget** of the empirical variogram $\hat{\gamma}(d)$ of station-mean SM, read off by extrapolating $d \rightarrow 0$. For CR200-18, $\hat{\gamma}(0.6 \text{ km}) \approx 0.0601^2 = 0.0036$. If the nugget is of that same order then $\text{SNR} \approx 1$ and $R_{\max}^2 \approx 0.5$: a probe scoring $r \approx 0.5$ would be *at the ceiling*, not failing.

This must be computed before the embedding sweep is interpreted. It costs minutes and uses labels only. Reporting a null without it would be uninterpretable.

4.4 The test statistic, and why it is legitimate at $n \ll p$

For random vectors $U \in \mathbb{R}^p$, $V \in \mathbb{R}^q$ with finite first moments, the distance correlation of Székely, Rizzo and Bakirov satisfies

$$\text{dCor}(U, V) = 0 \iff U \perp V. \quad (8)$$

Unlike Pearson correlation it detects *any* dependence, linear or not, and unlike mutual information it is estimable at small n . It has **no fitted parameters**, so overfitting is structurally impossible, and under H_0 the labels Δy are exchangeable, making the permutation null **exact** rather than asymptotic. With $B = 10^4$ permutations the p -value $(1 + \#\{\text{dCor}^* \geq \text{dCor}\})/(1 + B)$ is valid for any n and any p .

The one real hazard: distance concentration. In high dimension, pairwise Euclidean distances concentrate, $\text{Var}(\|\Delta e\|)/\mathbb{E}[\|\Delta e\|]^2 \rightarrow 0$ as $p \rightarrow \infty$, which attenuates dCor towards zero and costs power (it does not create false positives — the test stays valid, just conservative). Two mitigations, both leakage-free:

1. standardise each of the 768 coordinates using statistics computed on *far* pairs only;
2. project Δe onto the leading $k \approx 16$ principal components of the *far*-pair difference matrix. Far pairs never enter the near-pair test, so no information leaks across the hypothesis being tested.

4.5 The resolution curve, and its built-in positive control

Define, per separation bin,

$$\rho(d) := \text{dCor}(\Delta e, \Delta y \mid \|x_i - x_j\| \in d). \quad (9)$$

We already know ρ must be large at continental range: between-network level correlation is $r = +0.85$ (Table 1). Therefore **the far bins are the positive control**, computed with the identical statistic on the identical pipeline. No separately-constructed control is needed, and a pipeline bug that kills the signal would be caught where we *know* signal exists.

The separation at which $\rho(d)$ becomes indistinguishable from its permutation null is then a definition, not an inference: it **is** the effective level resolution of the embedding.

4.6 Arm A versus Arm C is a data-processing inequality

Pooling is a fixed, data-independent map P (`_cpu_pyramid_pool`, `dataset.py:190–220`) applied to the token grid. By the data-processing inequality, for any target y ,

$$I(P \cdot e; y) \leq I(e; y) \quad \implies \quad \rho_C \leq \rho_A \text{ in expectation.} \quad (10)$$

The comparison is therefore **one-sided by construction** — pooling can only destroy information, never create it. If $\rho_A > 0$ while $\rho_C \approx 0$, we have not merely observed a correlation: we have *proved* that the bottleneck is our pooling operator, not TerraMind.

4.7 Antisymmetry is a free guarantee

The labelling of i versus j is arbitrary, so the joint law of $(\Delta e, \Delta y)$ is invariant under $(\Delta e, \Delta y) \mapsto (-\Delta e, -\Delta y)$. Emitting both orderings of every pair makes this exact in the sample. Three consequences:

1. $\mathbb{E}[\Delta y] = \mathbb{E}[\Delta e] = 0$ *exactly*; the intercept is zero and must be fitted as zero (`fit_intercept=False`).
2. Any constant predictor achieves *exactly* the null RMSE. There is no free lunch from predicting the mean — the failure mode that makes ordinary station-level regression look better than it is.
3. The true regression function is odd, $f(-\Delta e) = -f(\Delta e)$. Ridge without intercept satisfies this automatically; the GBM is symmetrised by the augmentation.

4.8 Same-token pairs are an exact structural zero

A token covers 16×16 px = 160 m. Two stations closer than 160 m (and not straddling a token boundary) fall in the *same* token, so in Arm A

$$\Delta e \equiv 0 \quad \text{exactly, by construction,} \quad (11)$$

while $\Delta y \neq 0$ in general. These pairs are simultaneously:

- a **perfect negative control** — any nonzero apparent skill on them is a bug;
- a **direct estimate of the floor**: $\text{Var}(\Delta y \mid \text{same token}) = 2\gamma_\varepsilon(d_{<160\text{m}}) + 2\sigma_b^2$, which bounds σ_b^2 from above without any variogram extrapolation.

Per §26.10 this is not a hypothetical: of 95 temporally-usable in-tile pairs, roughly 62 are sub-token. They are not wasted — they measure the ceiling.

5 Design

5.1 Unit of analysis

For two stations i, j inside the same tile, per depth,

$$\Delta y_{ij} = \bar{S}M_i - \bar{S}M_j \quad \text{over their } \textit{common} \text{ observed dates (QC-clean only),} \quad \Delta e_{ij} = e_i - e_j.$$

Requiring common dates matters: differing record windows would reintroduce climate through the back door (a station observed only in a drought year is drier for reasons unrelated to place). §26.10 found temporal overlap, not geometry, is the binding constraint.

Station at pixel (r, c) lands in token $(\lfloor r/16 \rfloor, \lfloor c/16 \rfloor)$. The CR200-18 figure lists all six token indices — 105, 62, 100, 20, 44, 172 — so for that tile all six embeddings are genuinely distinct vectors.

5.2 The three arms

arm	embedding taken from	question it answers
A	both stations read from <i>one</i> tile’s 14×14 grid at their own token indices	does TerraMind’s representation distinguish two points 160–900 m apart <i>within one image</i> ?
B	each station uses the centre token (7, 7) of its <i>own</i> patch	is the finest thing the model is actually fed discriminative?
C	the 4-scale pooled pyramid the model really receives (<code>dataset.py:190–220</code>)	does the pooling destroy it?

Table 2: Arm A is what TerraMind knows; Arm C is what the model is told. The gap between them is our engineering, not TerraMind’s limit.

Arm B additionally unlocks *all* pairs at *any* separation — stations need not share a tile — which is what makes the $\rho(d)$ curve of §4.5 estimable with large n .

5.3 Layer × modality sweep

Verified on ISMN_TxSON_CR200-18 (read-only, 2026-08-12):

modality	arrays present	shape
<code>s2/{13,16,19,112}</code>	all four layers	(178, 196, 768) fp16
<code>s1_asc/..., s1_desc/...</code>	all four layers	(184, 196, 768) fp16
<code>dem, lulc</code>	L12 only	(196, 768) fp16

Sweep = $\{S2, S1_{ASC}, S1_{DESC}\} \times \{L3, L6, L9, L12\} + \{DEM, LULC\} \times \{L12\} = \mathbf{14 \text{ combinations}} \times 3 \text{ arms}$.

The hypothesis this sweep tests. L12 is semantic — it is where site identity lives, which §24.12 showed dominates the model’s behaviour. L3 and L6 are shallower and closer to raw radiometry, which is where a 160 m wetness contrast would live if it lives anywhere. **We predict early layers beat L12 on within-tile contrast even though L12 beats them on site identity.** A layer-monotone result either way is a real finding about what foundation-model features are for.

Temporal aggregation matches the target: Δy is a multi-year mean, so the primary feature is the multi-year mean token. Seasonal means and the temporal SD of the token enter as secondary blocks.

DEM and LULC at L3/L6/L9 do not exist and would require a GPU re-run of `precompute_terrmind.py` (one static forward per station, no time axis). Gated: do not pay for it unless the S2/S1 sweep shows early layers matter.

5.4 Statistical ladder and controls

Ordered least-overfittable first, per Figure 2. Controls:

Antisymmetry both orderings emitted; a constant predictor sits exactly at the null.

Label shuffle permute Δy within folds; every combination must collapse. Mandatory at $p = 768$, $n \sim 33$.

Same-token pairs $\Delta e \equiv 0$; any apparent skill is a bug (§4.8).

Far bins must be significant — the built-in positive control (§4.5).

Random vector a random $\mathbb{R}^{768}$ vector per station must give $dCor \approx 0$.

Independent positive control the same machinery on $\Delta(\text{elevation})$ and $\Delta(\text{soil texture})$ at the station pixel, which demonstrably vary in this tile (450.8–467.9 m; SOC 161–175). If nothing is detected there either, the machinery is broken, not TerraMind. This is the analogue of §24’s ERA5-shuffle control, which is what made §24 interpretable.

6 Data inventory

6.1 Pair counts — and the honest number

Great-circle sweep over all 993 stations in `csvs/station_splits.csv` (computed 2026-08-12):

separation	pairs	separation	pairs
< 0.5 km	46	2.24–5 km	178
0.5–1 km	23	5–10 km	283
1–2.24 km	57	10–30 km	1 420
< 2.24 km total	126	30–100 km	4 287
		> 100 km	486 234

The 126 span **12 networks** — TxSON 55, FMI 22, SNOTEL 13, AmeriFlux 11, SCAN 5, ICOS 4, ARM 3, FLUXNET-AMERIFLUX 3, SOILSCAPE 3, iRON 2, Berlin 1, NGARI 1 — so Arm A is not confined to one catchment or one climate.

But geometry is not the binding constraint. §26.10 already ran this inventory properly, including the 482 level-1 stations outside the pipeline, and found: 33 colocation clusters covering 105 stations; 206 ordered in-tile pairs; **95 usable at ≥ 365 d record overlap**; and restricting to ≥ 160 m separation — required for $\Delta e \neq 0$ in Arm A — leaves **33 unique pairs**. Sixty-four in-tile pairs are lost purely to non-overlapping records, including the 7-node SOILSCAPE deployment inside Tonzi Ranch’s tile.

So Arm A’s real sample size is $n \approx 33$ (66 with both orderings), not 126.

This is the single most important number in this document and it must be stated in any write-up. It is why the ladder begins with a parameter-free statistic and an exact permutation test, and why Arm B and the $\rho(d)$ curve — where n is in the thousands — carry the statistical weight.

6.2 Depth

0–10 cm is primary: it is where satellite sensing has physical justification and where the tile figure fails. 10–30 and 30–100 are reported as secondary. Note TxSON is **surface-only** (§26.2), so the deeper analysis draws on the non-TxSON clusters. §24.12 found deep prediction is close to

a pure site fingerprint (temporal share only 12 % at 30–100), so the depth dependence is itself informative.

6.3 Multiplicity

14 (modality $\times$ layer) $\times$ 3 arms $\times$ 3 depths $\times$ 8 distance bins. **Pre-registered primary test: depth 0–10, Arm A, S2, all bins.** Everything else is exploratory and reported under Benjamini–Hochberg FDR. The gate is *not* a max-over-tests.

7 How each outcome is read

Always jointly with the ceiling $R_{\max}^2(d)$ of §4.3 — a probe that reaches the ceiling has succeeded even if its absolute r looks modest.

Arm A	Arm C	conclusion	next move
signal	signal	tokens carry it, the model is fed it, the model fails to use it	training/loss problem: multi-station supervision, level/anomaly decomposition
signal	none	pooling destroys it — guaranteed one-sided by the DPI	cheapest fix in the project: restore a true 1×1 centre scale, exempt the centre token from the 50 % dropout
none	none, $R_{\max}^2 \gg 0$	TerraMind at 160 m does not resolve sub-km SM contrast	state the resolution limit; reframe on $\text{ubRMSE} / \text{NSE}_{\text{anom}}$
none	none, $R_{\max}^2 \approx 0$	the <i>observations</i> cannot resolve it — station offset noise exceeds the true field contrast	not a model finding at all; report the nugget and stop

Valid only if $\rho(d)$ is significant in the far bins and the random-vector control gives $\text{dCor} \approx 0$.

8 Limitations, stated up front

1. $n \approx 33$ **for Arm A.** A null is weak evidence of absence and must be reported as underpowered, never as “TerraMind cannot”. Power, not validity, is the constraint — the permutation test is exact at any n .
2. Δb **is irreducible.** Part of every observed contrast is instrument, not landscape. (7) quantifies how much; it cannot be removed.
3. **A 160 m token cannot be expected to resolve a point sensor.** ISMN probes integrate over centimetres. Even a perfect 160 m representation predicts the 160 m areal mean, not the probe. This is a fundamental support mismatch and it contributes to σ_b^2 .
4. **Arm A tests the representation, not the model.** A positive Arm A does not imply the current architecture could exploit it; that is what Arm C and the deferred multi-station-supervision work would establish.
5. **Colocated stations are geographically clustered.** Even across 12 networks, clusters are not a random sample of the Earth. Conclusions generalise to “dense-network conditions”, not globally.
6. **Mean SM is one summary.** Two stations could share a mean and differ in dynamic range. This experiment deliberately targets level, because level is where the model fails ($\text{bias}^2 \approx 50\%$ of MSE per §22.10).

9 What gets built, after this document is approved

- `build_network_readouts.py` — add `--all-networks`, writing `csvs/all_readouts.csv`; reuse the existing centre assertion (:214) and forward/reverse offset symmetry check (:230). Must reproduce `csvs/txson_readouts.csv` when filtered to TxSON.
- `probe_variogram.py` — §4.3, labels only, minutes. Produces $\hat{\gamma}(d)$, the nugget, and $R_{\max}^2(d)$ per depth. **Runs first.**
- `probe_terramind_subkm.py` — the sweep. Modelled on `station_mean_probe.py`: CPU-only, `Pool(64)`, opens `zarr` per station directly (never instantiates `SoilMoistureDataset`, which eagerly loads ~ 16 GB of L12 tokens). Reuses `_load_zarr_labels` and `_cpu_pyramid_pool` from `dataset.py`.
- `slurm/probe_terramind_subkm.sh` — CPU partition, `--cpus-per-task=64`, `--mail-type=BEGIN,END,FAIL`, `--mail-user=ktm.prajwalkhanal@gmail.com`. Nothing runs interactively.
- Outputs: `csvs/subkm_pairs.csv`, `csvs/terramind_subkm_probe.json`, `figures/subkm_layer_sweep.png` (dCor heat-map, layer $\times$ modality $\times$ arm), `figures/subkm_rho_curve.png` (the $\rho(d)$ resolution curve).

Verification before any number is trusted. (i) every station projects into its own tile at exactly (112,112); (ii) the six CR200-18 station means reproduce the figure’s values (0.1367, 0.1197, 0.1826, 0.2323, 0.1857, 0.2865) to 10^{-3} ; (iii) for a tile’s centre station, its Arm A token and Arm B token are bit-identical; (iv) $\|\Delta e\| = 0$ exactly on same-token pairs; (v) $\sum \Delta y = 0$ exactly over the antisymmetrised table; (vi) all 14 combinations collapse to the null under permuted labels.